Completing the Square

Name _____ **Date** _____ No calculator Version 1

KEY WORDS perfect square · completing the square · quadratic · surd · exact answer · minimum

A Perfect squares ●○○

ANSWER ON THE SHEET

Remember: $(x+p)^2 = x^2 + 2px + p^2$. The number in front of x is doubled, and the last number is squared.

1. Expand.

a) $(x+3)^2$ = _____ **b)** $(x-5)^2$ = _____ **c)** $(x+7)^2$ = _____

d) $(x-1)^2$ = _____ **e)** $(x+10)^2$ = _____ **f)** $(x-6)^2$ = _____

2. Fill in the boxes to make each one a perfect square.

a) $x^2 + 6x + \square = (x + \square)^2$ **b)** $x^2 + 10x + \square = (x + \square)^2$

c) $x^2 - 8x + \square = (x - \square)^2$ **d)** $x^2 - 14x + \square = (x - \square)^2$

e) $x^2 + 20x + \square = (x + \square)^2$ **f)** $x^2 - 18x + \square = (x - \square)^2$

g) $x^2 + 2x + \square = (x + \square)^2$ **h)** $x^2 - 24x + \square = (x - \square)^2$

B Completed-square form ●●○

WORK IN YOUR BOOK

Remember: Halve the number in front of x to get the bracket. Then subtract the square of that number.

3. Write each expression in the form $(x+p)^2 + q$.

a) $x^2 + 6x + 2$ **b)** $x^2 + 4x + 9$ **c)** $x^2 + 8x + 5$

d) $x^2 - 2x + 7$ **e)** $x^2 - 6x + 1$ **f)** $x^2 + 10x + 30$

g) $x^2 - 4x - 3$ **h)** $x^2 + 2x - 8$ **i)** $x^2 - 8x + 20$

j) $x^2 + 12x$ **k)** $x^2 - 10x + 11$ **l)** $x^2 + 14x + 50$

C Solving when the square is complete ●●○

WORK IN YOUR BOOK

Remember: Take the square root of both sides. A positive number has two square roots, so write $\pm$.

4. Solve. Leave any square roots in your answers.

a) $(x+3)^2 = 16$ **b)** $(x-2)^2 = 25$ **c)** $(x+5)^2 = 1$

d) $(x-4)^2 = 49$ **e)** $(x+1)^2 = 5$ **f)** $(x-3)^2 = 7$

g) $(x+2)^2 - 9 = 0$ **h)** $(x-6)^2 - 4 = 0$ **i)** $(x+4)^2 - 12 = 0$

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D Solving by completing the square ●●● WORK IN YOUR BOOK

⚠ **Watch out:** When you take the square root of both sides, remember the $\pm$.

5. Solve by completing the square. Leave any square roots in your answers.

a) $x^2 + 6x + 5 = 0$ **b)** $x^2 + 4x - 5 = 0$ **c)** $x^2 - 8x + 12 = 0$

d) $x^2 - 2x - 15 = 0$ **e)** $x^2 + 10x + 9 = 0$ **f)** $x^2 - 12x + 20 = 0$

g) $x^2 + 4x - 1 = 0$ **h)** $x^2 + 6x + 2 = 0$ **i)** $x^2 - 2x - 4 = 0$

j) $x^2 - 10x + 20 = 0$ **k)** $x^2 + 8x + 4 = 0$ **l)** $x^2 - 4x - 8 = 0$

E The smallest value ●●● WORK IN YOUR BOOK

Remember: A square is never negative, so $(x - 3)^2 + 2$ is smallest when the bracket is 0. Its smallest value is 2, when $x = 3$.

6. Complete the square. Then write down the smallest value of the expression and the value of x that gives it.

a) $x^2 - 6x + 11$ **b)** $x^2 + 4x + 7$

c) $x^2 - 2x + 5$ **d)** $x^2 + 8x + 10$

e) $x^2 - 10x + 21$ **f)** $x^2 + 6x - 1$

g) $x^2 - 12x + 40$ **h)** $x^2 + 2x - 6$

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7. Write $x^2 + 5x + 1$ in the form $(x + p)^2 + q$. Here p and q are fractions.

8. Write $2x^2 + 8x + 3$ in the form $a(x + p)^2 + q$.

9. Solve $x^2 + 3x - 1 = 0$ by completing the square. Give exact answers.

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Completing the Square

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS perfect square · completing the square · quadratic · surd · exact answer · minimum

A Perfect squares ●○○ — ANSWER ON THE SHEET

Remember: $(x+p)^2 = x^2 + 2px + p^2$. The number in front of x is doubled, and the last number is squared.

1. Expand.

a) $(x+4)^2$ = _____ **b)** $(x-3)^2$ = _____ **c)** $(x+8)^2$ = _____

d) $(x-2)^2$ = _____ **e)** $(x+9)^2$ = _____ **f)** $(x-7)^2$ = _____

2. Fill in the boxes to make each one a perfect square.

a) $x^2 + 8x + \square = (x + \square)^2$ **b)** $x^2 + 12x + \square = (x + \square)^2$

c) $x^2 - 6x + \square = (x - \square)^2$ **d)** $x^2 - 16x + \square = (x - \square)^2$

e) $x^2 + 18x + \square = (x + \square)^2$ **f)** $x^2 - 20x + \square = (x - \square)^2$

g) $x^2 + 4x + \square = (x + \square)^2$ **h)** $x^2 - 22x + \square = (x - \square)^2$

B Completed-square form ●●○ — WORK IN YOUR BOOK

Remember: Halve the number in front of x to get the bracket. Then subtract the square of that number.

3. Write each expression in the form $(x+p)^2 + q$.

a) $x^2 + 4x + 1$ **b)** $x^2 + 6x + 13$ **c)** $x^2 + 10x + 3$

d) $x^2 - 2x + 9$ **e)** $x^2 - 8x + 5$ **f)** $x^2 + 8x + 21$

g) $x^2 - 6x - 2$ **h)** $x^2 + 2x - 5$ **i)** $x^2 - 4x + 12$

j) $x^2 + 10x$ **k)** $x^2 - 12x + 16$ **l)** $x^2 + 16x + 70$

C Solving when the square is complete ●●○ — WORK IN YOUR BOOK

Remember: Take the square root of both sides. A positive number has two square roots, so write $\pm$.

4. Solve. Leave any square roots in your answers.

a) $(x+2)^2 = 9$ **b)** $(x-3)^2 = 36$ **c)** $(x+6)^2 = 1$

d) $(x-5)^2 = 64$ **e)** $(x+1)^2 = 3$ **f)** $(x-2)^2 = 11$

g) $(x+3)^2 - 16 = 0$ **h)** $(x-7)^2 - 4 = 0$ **i)** $(x+5)^2 - 18 = 0$

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D Solving by completing the square ●●● WORK IN YOUR BOOK

⚠ **Watch out:** When you take the square root of both sides, remember the $\pm$.

5. Solve by completing the square. Leave any square roots in your answers.

a) $x^2 + 6x + 8 = 0$ **b)** $x^2 + 4x - 12 = 0$ **c)** $x^2 - 8x + 7 = 0$

d) $x^2 - 2x - 8 = 0$ **e)** $x^2 + 10x + 16 = 0$ **f)** $x^2 - 12x + 11 = 0$

g) $x^2 + 4x + 1 = 0$ **h)** $x^2 + 6x + 4 = 0$ **i)** $x^2 - 2x - 6 = 0$

j) $x^2 - 10x + 18 = 0$ **k)** $x^2 + 8x + 8 = 0$ **l)** $x^2 - 4x - 4 = 0$

E The smallest value ●●● WORK IN YOUR BOOK

Remember: A square is never negative, so $(x - 3)^2 + 2$ is smallest when the bracket is 0. Its smallest value is 2, when $x = 3$.

6. Complete the square. Then write down the smallest value of the expression and the value of x that gives it.

a) $x^2 - 4x + 9$ **b)** $x^2 + 6x + 12$

c) $x^2 - 2x + 8$ **d)** $x^2 + 10x + 20$

e) $x^2 - 8x + 19$ **f)** $x^2 + 4x - 3$

g) $x^2 - 14x + 50$ **h)** $x^2 + 2x - 9$

★ Extension FOR EARLY FINISHERS

7. Write $x^2 + 3x + 4$ in the form $(x + p)^2 + q$. Here p and q are fractions.

8. Write $3x^2 - 12x + 5$ in the form $a(x + p)^2 + q$.

9. Solve $x^2 + 5x - 1 = 0$ by completing the square. Give exact answers.

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